

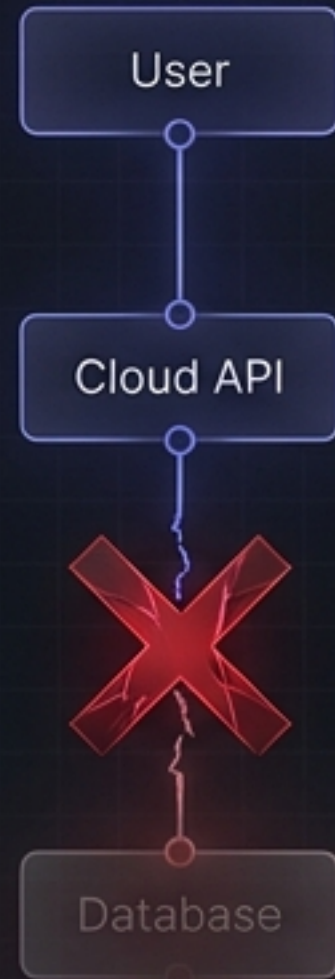
Engineering AgendOS

VocalPlan: The
intelligent calendar
operating system.

Built on React 19,
TypeScript, and
Capacitor 8.

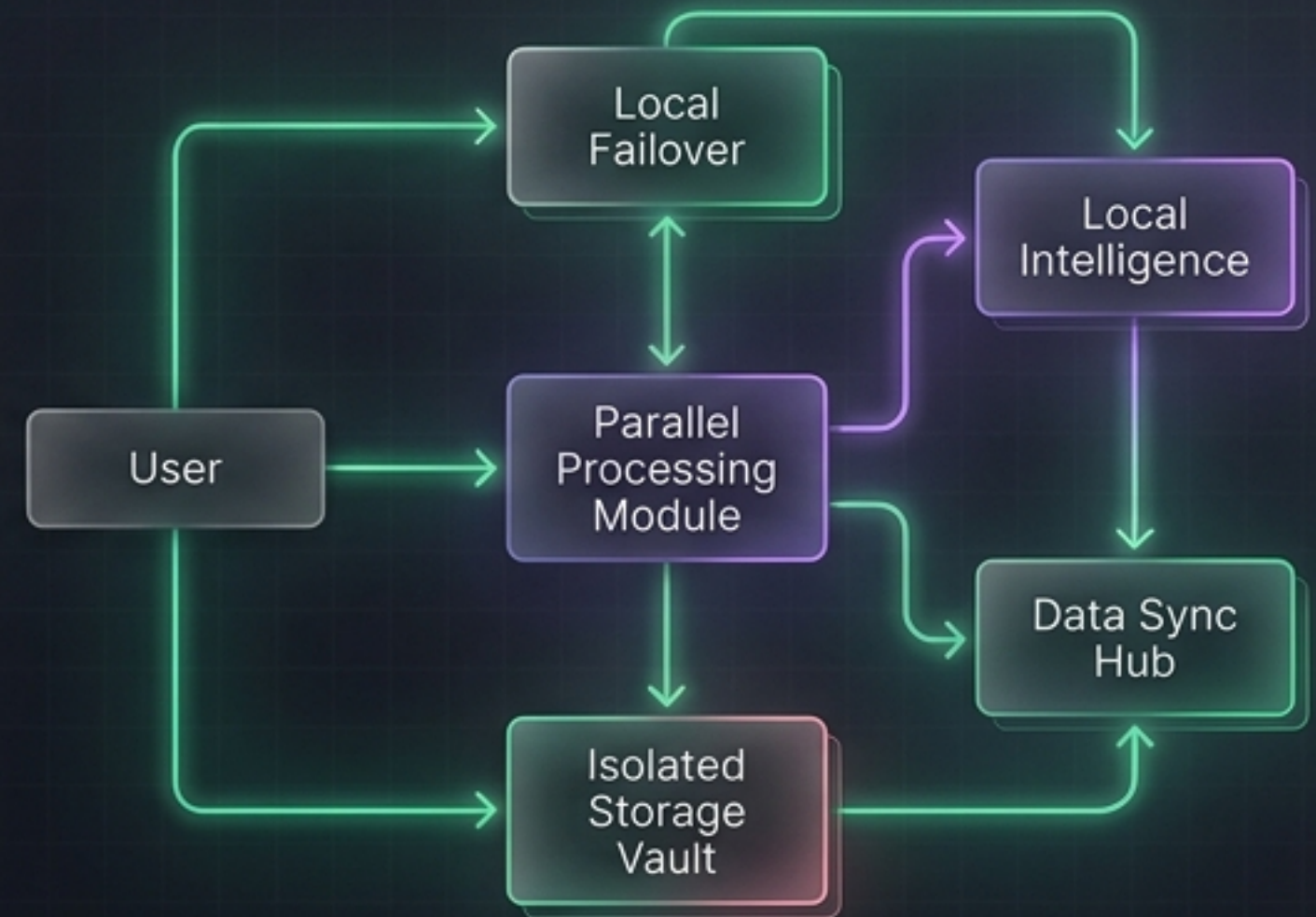
Architected for
zero-latency UI,
offline reliability,
and multimodal
intelligence.

The Wrapper



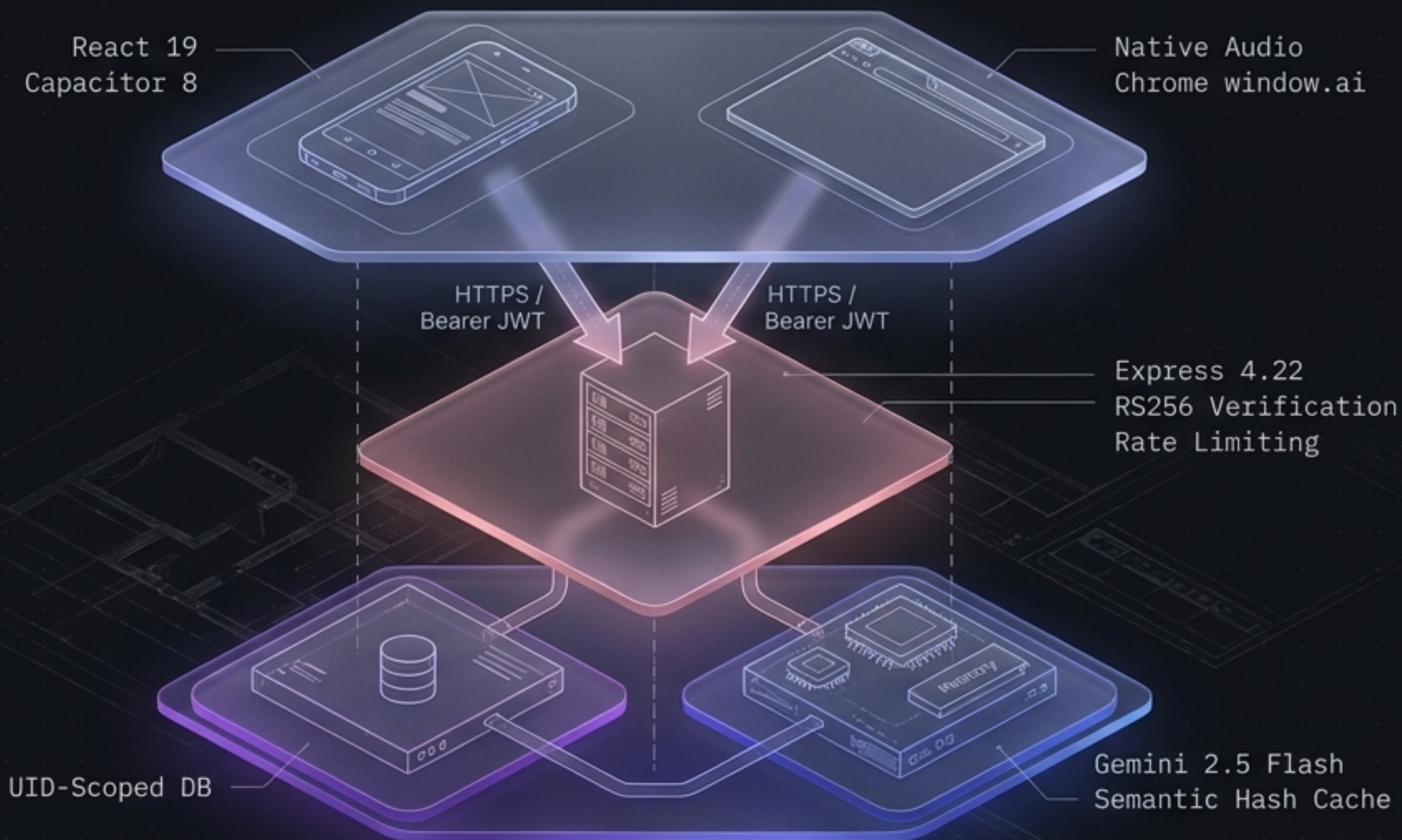
Cloud-dependent AI scheduling suffers from latency, offline failure, and data privacy vulnerabilities.

The AgendOS Solution



Push intelligence to the edge, treat the network as an enhancement rather than a dependency, and adopt a zero-trust security posture.

The VocalPlan system topology



Three foundational pillars of architecture



I. Multimodal Intelligence

Hybrid AI routing, semantic caching, and hands-free accessibility.



II. Local-First Reliability

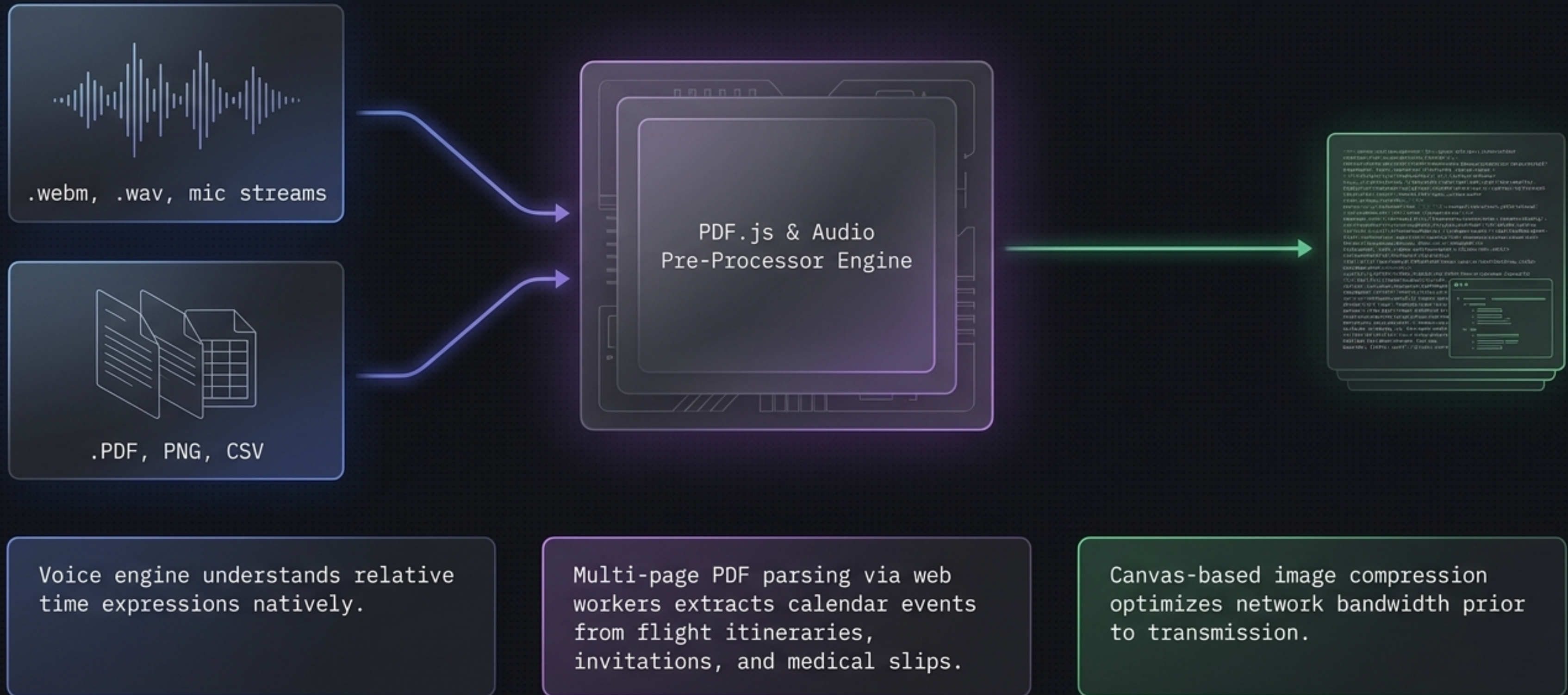
Optimistic UI, offline synchronization queues, and deterministic edge rendering.



III. Zero-Trust Security

Cryptographic JWT validation, strict data isolation, and robust sanitization.

Multimodal ingestion handles continuous inputs

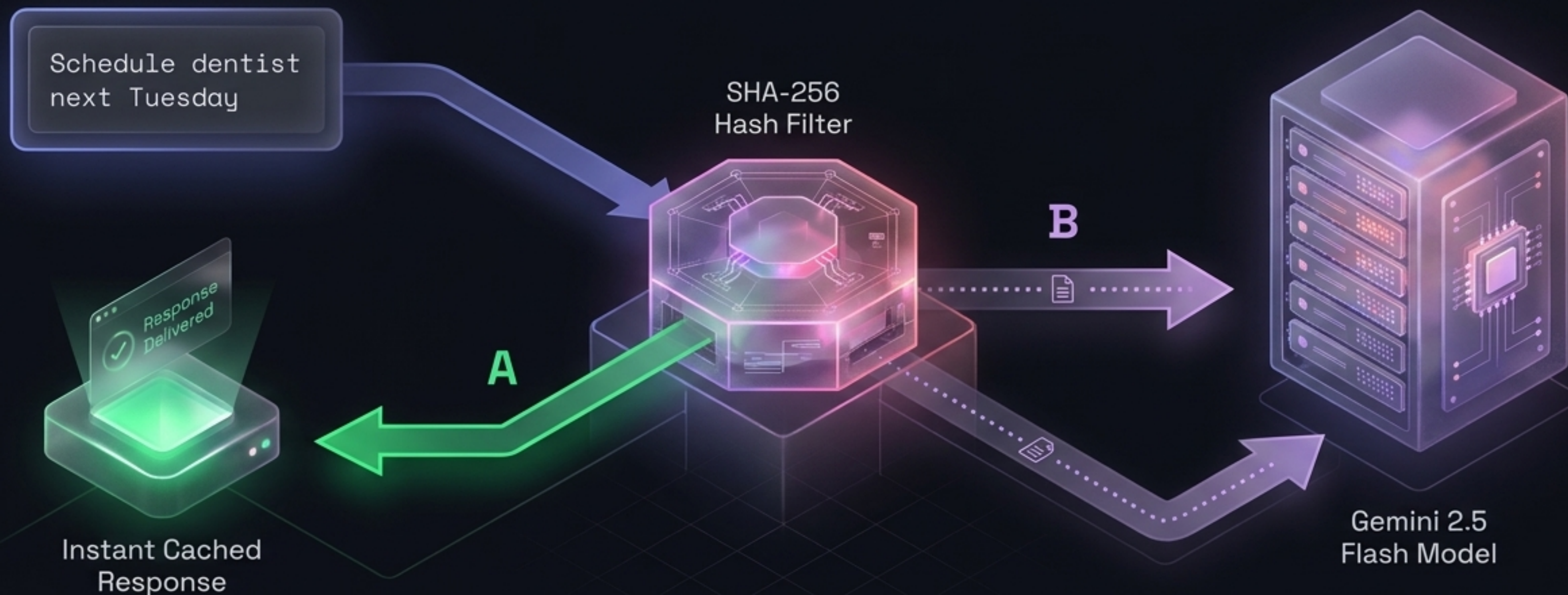


Hybrid intelligence routing matrix

The Routing Matrix

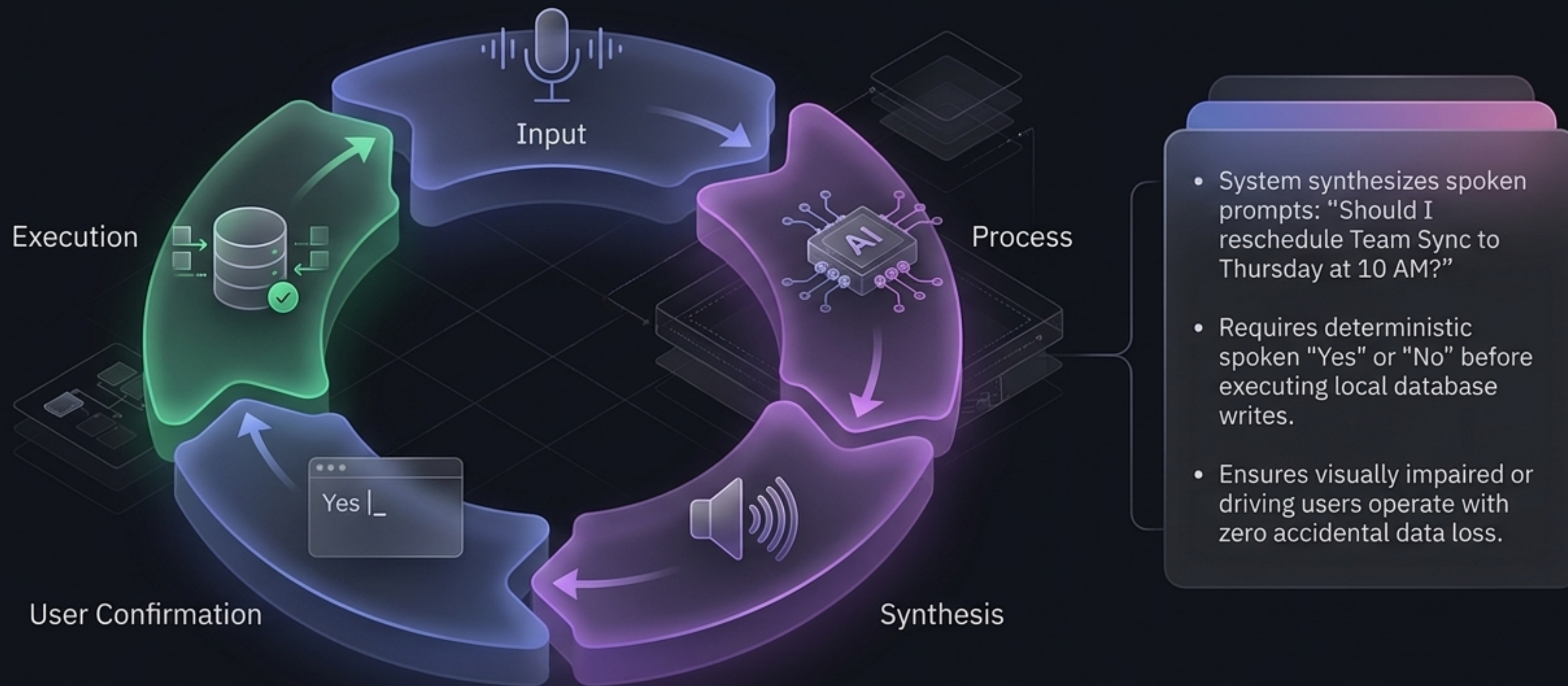
Path	Environment	Latency	Use-Case	Cost/Privacy
Android AICore	On-Device Mobile	Zero	Supported Hardware	Free / Private
Chrome window.ai	On-Device Web	Zero	Web Nano	Free / Private
Deterministic NLP	Client-Side Code	Sub-millisecond	Basic Text	Free / Private
Cloud Gemini 2.5 Flash	Secure Proxy	Network-dependent	Complex Multimodal	API Cost

Semantic Hash Caching intercepts redundant queries



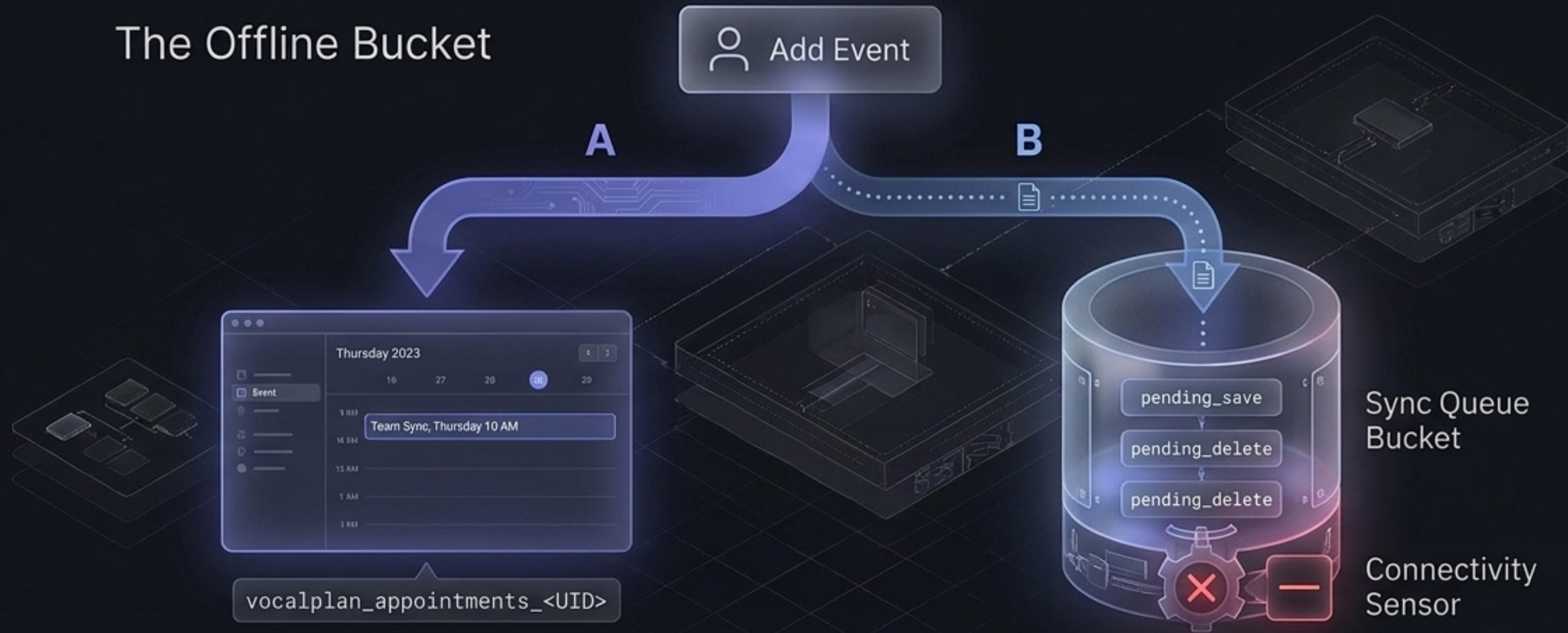
Computes SHA-256 hashes of queries and multimodal inputs to return instant cached responses, bypassing API consumption entirely.

The accessibility and verbal confirmation loop



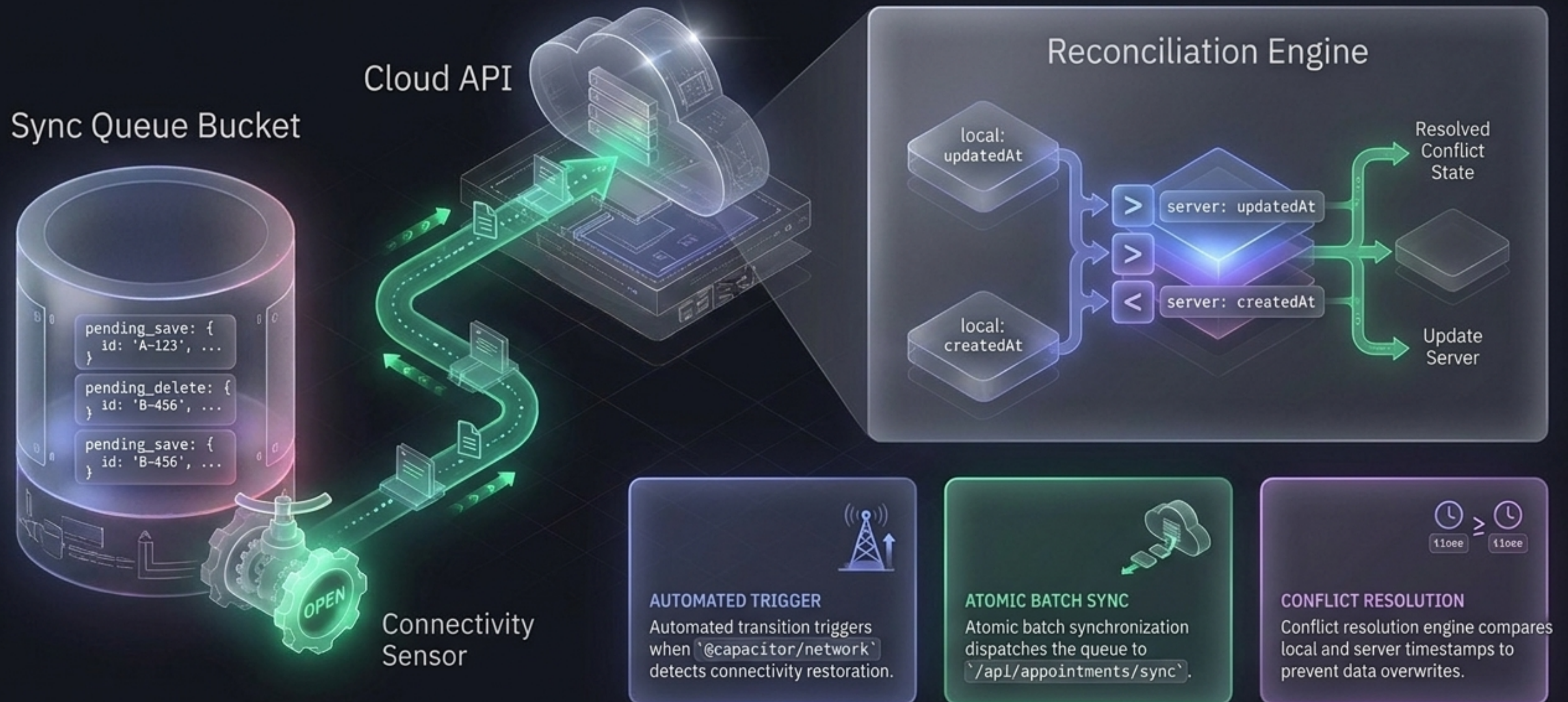
Local-first architecture guarantees zero latency

The Offline Bucket

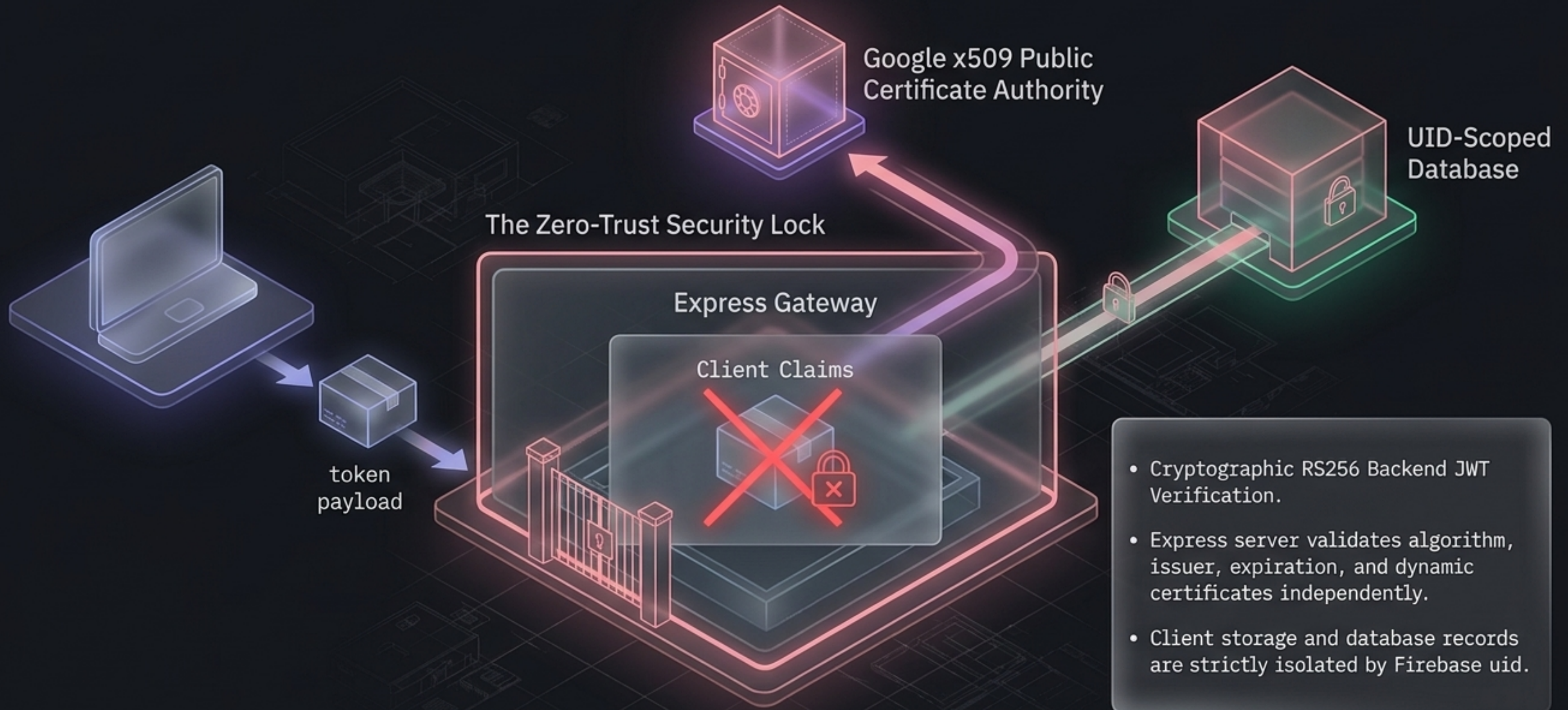


- Immediate optimistic UI execution via IndexedDB and scoped localStorage.
- Network monitor (@capacitor/network) holds mutations as pending_save or pending_delete during offline states.

Atomic synchronization and conflict reconciliation



Enforcing a strict zero-trust client model



Defense-in-depth across the application stack

Prototype Pollution / Object Injection

`sanitizeAppointment()` strips `__proto__` and enforces a strict 20-field whitelist.

Cross-Site Scripting (XSS)

Helmet Content Security Policy enforces `script-src 'self'`.

Denial of Service (DoS)

Dual-tier rate limiting (600 req/15min general; 45 req/min AI endpoints).

Filters threshold
Layer 1P threshold

In-App Purchase Replay Attacks

Server computes SHA-256 hashes of Play Store receipt tokens to block reuse across multiple accounts.

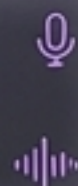
token

hash

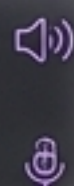
The full capability grid of AgendOS

Voice & AI

Multimodal Scheduler



Verbal Confirmation



Document Scanner



Smart Title Suggestion

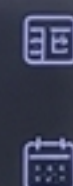


Hybrid AI Monitor

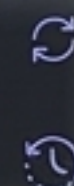


Calendar

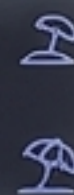
Multi-Perspective Views



Recurrence Engine



Vacation Spans



National Holidays



Color Categorization

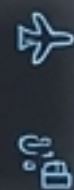


Storage & Sync

Local-First Core



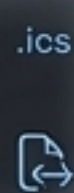
Offline Queue



Soft-Delete Recycle Bin

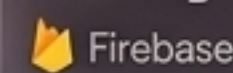


.ics/.csv Import & Export

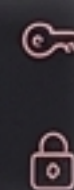


Security & System

Firebase Identity



Zero-Trust RS256



Weather Integration



Exact Alarms



Multilingual Localization



• User speaks a command.

• Connectivity returns.

Pillar I (Intelligence):
Processed locally via
device NLP/Nano.

Pillar II (Reliability):
Saved instantly to
Optimistic UI/IndexedDB.

Pillar III (Security):
Payload cryptographically
signed and synced to the
secure backend.

Multimodal Intelligence + Local-First Reliability + Zero-Trust Security

Complex deterministic architecture designed to deliver an invisible, flawless user experience under worst-case conditions

The architectural roadmap for VocalPlan

